PHOTOVOLTAIC SYSTEM

PRELIMINARY DESIGN DRAWINGS

PROJECT	**Ground 13kWp 3-phase**
CLIENT	**Test Client**
SITE	**Plot 12, Mbabane, Eswatini**
DRAWN BY	**N. Hlophe**
DATE / REVISION	**2026-11-12 / A**

DESIGN SUMMARY

Array	24 x 550 W = 13.2 kWp
Modules	Generic 550 W mono PERC
Inverter	Generic 10 kW hybrid, 3-phase
DC/AC ratio	1.32
Strings	2 x 12 modules
String voltage	Voc cold 637.7 V, Vmp 501.6 V (limit 1000 V)
AC protection	20 A 4P MCB, 30 mA RCD
Battery	15 kWh, 400 V
Mounting	Ground mount (tilted frames), 26 deg, 12 x 2

DRAWING INDEX

PV-001-01	Cover sheet and design summary
PV-001-02	Single line diagram
PV-001-03	Array layout plan
PV-001-04	Mounting structure detail
PV-001-05	Bill of materials and schedules

DESIGN CHECKS

No warnings raised by the checks.

IMPORTANT

These drawings are generated automatically from the inputs shown and are PRELIMINARY.

They are a design aid, not certified or stamped designs, and must not be used for construction until checked and signed off by a registered engineer or competent electrician.

Module and inverter values are taken from user input or generic defaults; replace them with datasheet values.

Compliance with SANS 10142-1, IEC 60364-7-712, IEC 62548, SANS 10160 and the local utility (EEC) requirements must be confirmed. Structural loading (wind, snow, foundations) is not calculated.

Designed by	Checked by	Approved (registered engineer)

Solar & Electrical Design AI

Project: Ground 13kWp 3-phase Client: Test Client

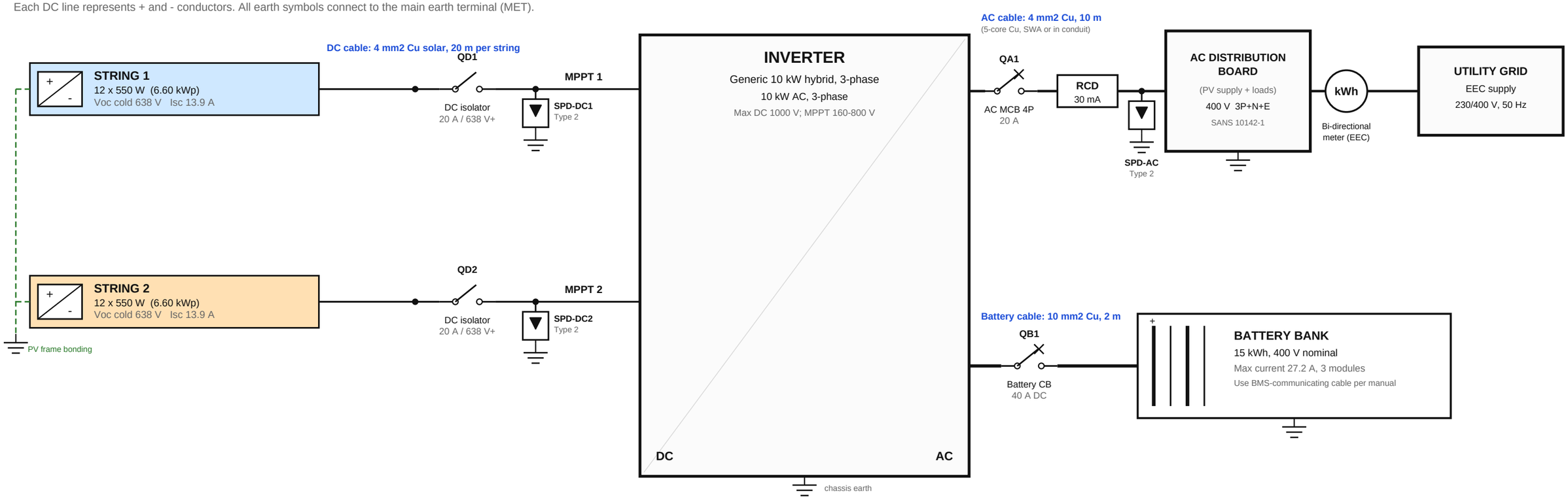
Plot 12, Mbabane, Eswatini

TITLE	DWG PV-001-01	SHEET 1 / 5
COVER SHEET AND DESIGN SUMMARY	REV A 2026-11-12	SCALE NTS

PRELIMINARY - NOT FOR CONSTRUCTION - subject to review by a competent person

SINGLE LINE DIAGRAM

Each DC line represents + and - conductors. All earth symbols connect to the main earth terminal (MET).



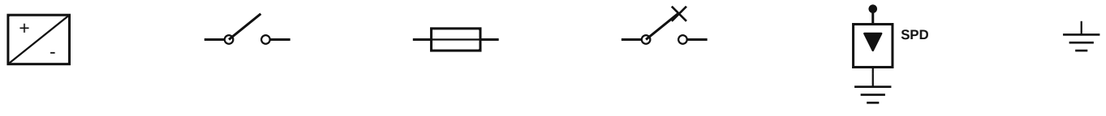
STRING AND MPPT DATA

Input	Strings	Mod/string	Voc cold V	Vmp V	Isc A	Imp A	Limit Iop/Isc A	Check
MPPT 1	1	12	638	502	13.9	13.2	26 / 34	OK
MPPT 2	1	12	638	502	13.9	13.2	26 / 34	OK

GENERAL NOTES

1. DC cables: double-insulated PV cable, UV resistant, routed in conduit or trunking.
2. Label all isolators, breakers and the DC/AC warning signs per SANS 10142-1.
3. Install AC and DC SPDs as close as practical to the inverter.
4. Fit a lockable AC isolator for the utility if the EEC requires it.
5. Confirm connection approval and metering requirements with EEC before installing.

LEGEND



PV string · DC isolator · Fuse (gPV) · Breaker · SPD · Earth

Blue text = cable. Dashed green = earth bonding.

Solar & Electrical Design AI

Project: Ground 13kWp 3-phase Client: Test Client

Plot 12, Mbabane, Eswatini

TITLE

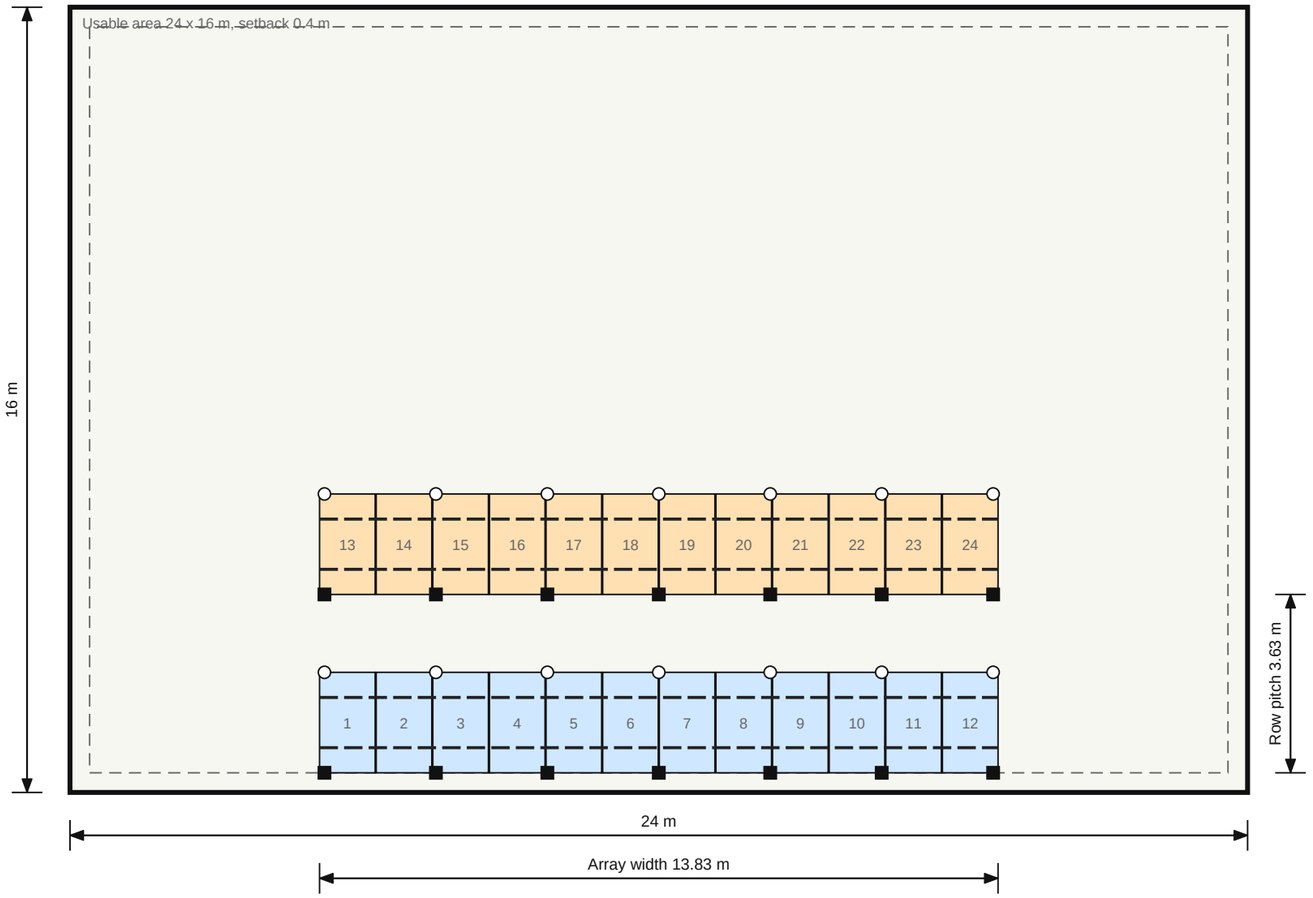
SINGLE LINE DIAGRAM

PRELIMINARY - NOT FOR CONSTRUCTION - subject to review by a competent person

DWG PV-001-02	SHEET 2 / 5
REV A 2026-11-12	SCALE NTS

ARRAY LAYOUT - GROUND MOUNT PLAN

Ground mount (tilted frames). Modules coloured by string. Dashed lines = rails. Plan view; arrays face down the sheet.



Modules face down the sheet

LEGEND

- String 1 (12 modules)
- String 2 (12 modules)
- Mounting rail
- Front post / foundation
- Rear post / foundation

QUANTITIES

Modules placed: 24
Layout: 12 per row x 2 row(s)
Orientation: portrait
Rail pieces: 16 x 4.2 m
Splices: 12 End clamps: 8
Mid clamps: 44
Tilt: 26 deg

Row pitch 3.63 m keeps rows shade-free from about 9:00 to 15:00 solar time on the winter solstice.

Wind uplift and foundations are not calculated. A structural engineer must size posts, foundations or ballast (SANS 10160-3 wind actions).

Solar & Electrical Design AI

Project: Ground 13kWp 3-phase Client: Test Client

Plot 12, Mbabane, Eswatini

TITLE	DWG PV-001-03	SHEET 3 / 5
ARRAY LAYOUT PLAN	REV A 2026-11-12	SCALE 1:125 (A3)

PRELIMINARY - NOT FOR CONSTRUCTION - subject to review by a competent person

MOUNTING STRUCTURE DETAIL

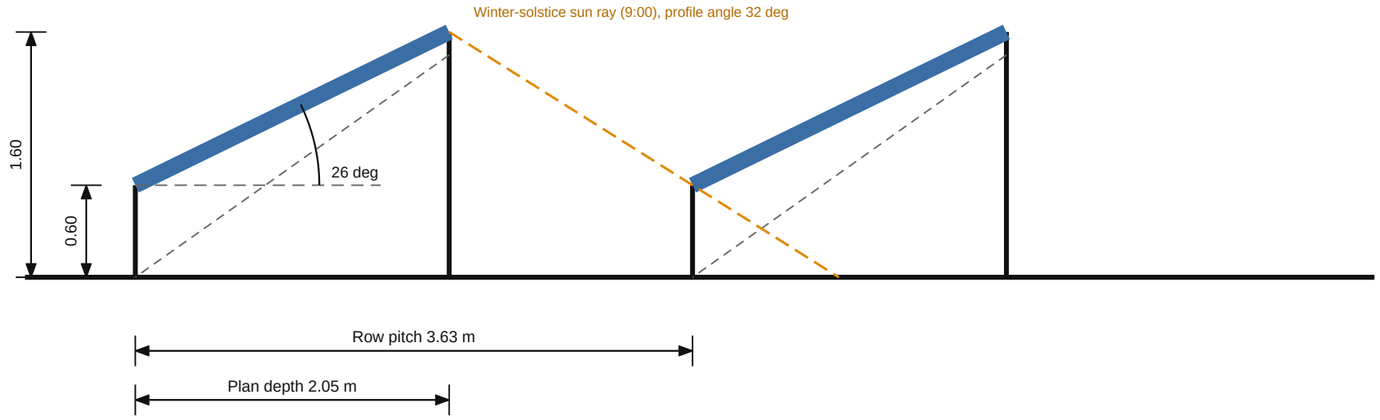
Ground mount (tilted frames). Side and front views to scale; clamp detail not to scale.

TILTED FRAMES - GROUND MOUNT

1 module(s) high, tilt 26 deg, front clearance 0.60 m, rear height 1.60 m. Section at 1:40.

Row pitch 3.63 m (ground cover ratio 0.56) keeps rows clear of shade from about 9:00 to 15:00 solar time on the winter solstice.

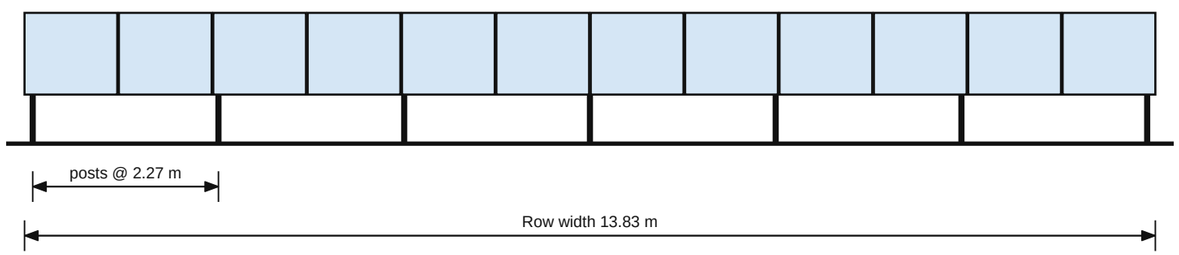
Foundations, post sizes and ballast: by structural engineer (wind uplift is not calculated).



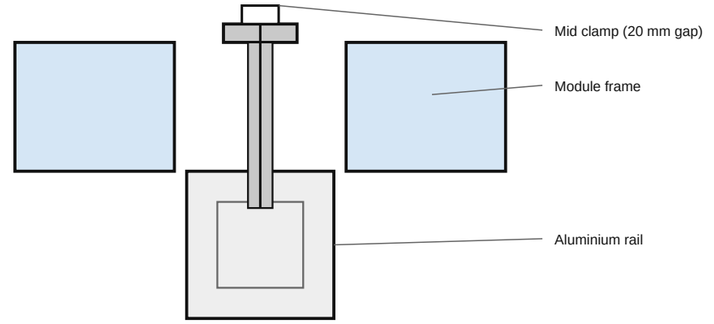
<- FRONT: modules face this way

FRONT ELEVATION - FRONT ROW (seen from the front, to scale)

1:75



TYPICAL MID CLAMP AND RAIL DETAIL (NTS)



Torque, clamp zone and frame depth per module and clamp manufacturer.

NOTES

1. Rail positions are generic (about 1/4 and 3/4 of the module). Use the clamping zone in the module manual.
2. Bond all rails, clamps and module frames; bond across rail splices.
3. Use stainless fasteners (A2/A4); avoid contact between dissimilar metals.
4. Check roof or ground structure capacity before installation. Wind and snow loads per SANS 10160.

Solar & Electrical Design AI

Project: Ground 13kWp 3-phase Client: Test Client

Plot 12, Mbabane, Eswatini

TITLE	DWG PV-001-04	SHEET 4 / 5
MOUNTING STRUCTURE DETAIL	REV A 2026-11-12	SCALE see views

PRELIMINARY - NOT FOR CONSTRUCTION - subject to review by a competent person

BILL OF MATERIALS AND SCHEDULES

Quantities are preliminary take-offs. Add wastage and confirm against final product choice.

No	Category	Description	Specification	Qty	Unit
1	Generation	Generic 550 W mono PERC	550 W, Voc 49.9 V, Isc 13.9 A, 2278x1134 mm	24	pcs
2	Conversion	Generic 10 kW hybrid, 3-phase	10 kW, 3-phase, 2 MPPT	1	pcs
3	Storage	Battery bank 15.0 kWh	400 V nominal LiFePO4	3	pcs
4	Protection	Battery DC breaker / fuse	40 A DC	1	pcs
5	Protection	DC isolator, MPPT 1	20 A, >= 638 V DC	1	pcs
6	Protection	DC isolator, MPPT 2	20 A, >= 638 V DC	1	pcs
7	Protection	DC surge protection	Type 2, Uc >= 1000 V DC	1	set
8	Protection	AC main breaker	20 A, 4P	1	pcs
9	Protection	RCD / RCBO	30 mA (type per inverter manual)	1	pcs
10	Protection	AC surge protection	Type 2	1	set
11	Cabling	Solar cable (DC strings)	4 mm2	88	m
12	Cabling	AC cable inverter to DB	4 mm2	11	m
13	Cabling	Battery cable	10 mm2	4.4	m
14	Cabling	MC4-type connector pairs	Same make as the modules	6	pairs
15	Earthing	Earth bonding cable	6 mm2 Cu green/yellow (16 mm2 if unprotected)	39	m
16	Earthing	Earth rod / bar and clamps		1	set
17	Other	Labels, warning signs and as-built folder		1	set
18	Mounting	Aluminium mounting rail	4.2 m lengths, anodised (55.3 m needed before offcuts)	16	pcs
19	Mounting	Rail splice		12	pcs
20	Mounting	End clamp	to suit 35 mm frame	8	pcs
21	Mounting	Mid clamp	20 mm gap	44	pcs
22	Mounting	Front post	0.60 m above ground	14	pcs
23	Mounting	Rear post	1.60 m above ground	14	pcs
24	Mounting	Inclined rafter (frame leg to leg)	2.38 m, 26 deg	14	pcs
25	Mounting	Diagonal brace	one per frame	14	pcs
26	Mounting	Foundation (pile / ground screw / concrete footing)	size and depth to be set by soil/wind design	28	pcs
27	Mounting	Earthing lug / bonding jumper		16	pcs

CABLE SCHEDULE

Tag	Circuit	Size	Length	I design	V drop
C1	DC strings (x2)	4 mm2	20 m	17.4 A	0.78 %
C2	AC inverter-DB	4 mm2	10 m	14.4 A	<= 3 %
C3	Battery	10 mm2	2 m	27.2 A	<= 1 %
E1	PV frame bonding	6 / 16 mm2	-	-	-

Sizes use a voltage-drop check plus a conservative current rating.
Verify installation method, grouping, ambient temperature.

PROTECTION SCHEDULE

Tag	Device	Rating
QD1	DC isolator	20 A, 638 V DC min
SPD-DC1	DC surge protector	Type 2, Uc >= 1000 V
QD2	DC isolator	20 A, 638 V DC min
SPD-DC2	DC surge protector	Type 2, Uc >= 1000 V
QA1	AC MCB 4P	20 A, curve C
RCD1	Residual current dev.	30 mA (type per inverter)
SPD-AC	AC surge protector	Type 2
QB1	Battery DC breaker	40 A DC

Solar & Electrical Design AI

Project: Ground 13kWp 3-phase Client: Test Client

Plot 12, Mbabane, Eswatini

TITLE

BILL OF MATERIALS AND SCHEDULES

PRELIMINARY - NOT FOR CONSTRUCTION - subject to review by a competent person

DWG PV-001-05

REV A 2026-11-12

SHEET 5 / 5

SCALE NTS